Sports Planning Assistant Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2025-26

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Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement & Objectives

Problem Statement

Planning sports activities, training schedules, and team management is often done manually, which leads to inefficiencies, poor resource utilization, scheduling conflicts, and lack of data-driven decisions. Coaches, players, and sports organizers struggle to optimize performance due to the absence of intelligent tools that can analyze data and provide actionable insights

Project Objectives

The main objective of the Sports Planning Assistant Agent is to develop an intelligent system that helps in planning, scheduling, and optimizing sports activities. The system aims to:

- Automate sports scheduling and planning
- Provide data-driven recommendations for training and performance
- Assist coaches in decision-making
- Improve team coordination and resource management
- Analyze player performance and suggest improvement

Scope of the Project

- Automates sports scheduling for matches, training sessions, and events.
- Manages player information and performance data in a structured format.
- Provides data analysis and insights to support decision-making.
- Generates optimized schedules based on availability and constraints.
- Displays visual dashboards and reports for better understanding of performance.
- Designed for schools, colleges, and small sports organizations.
- Acts as a decision-support system to improve planning and resource utilization.

Out-of-Scope

- The system does not support **real-time tracking or live match data integration**.
- It does not include **advanced AI/ML models for predictive analytics or injury prediction**.
- The project is not designed for **large-scale professional or multi-organization deployment**..

Proposed Solution

The proposed solution is a **two-layer intelligent sports planning system** that converts structured inputs (player data, schedules, performance metrics) into optimized planning outputs.

1. Presentation Layer (Frontend)

- Developed using a web-based interface (Streamlit), it provides:
- Dashboard-style user interface
- Player and schedule input forms
- Planning and recommendation modules
- Visualization panels for insights
- Output display for schedules and reports

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2. Application & Analytics Layer (Backend)

- Developed using Python, it includes:
- Input Validation Module
  Ensures correct format of player and schedule data
- Data Processing Module
  Cleans and structures input data
- Analytics & Optimization Module
  Applies logic to generate optimized schedules
- Recommendation Engine
  Suggests improvements in training and planning
- Visualization Module
  Generates charts and reports
- Error Handling Module
  Ensures system reliability

Key Features

1. Automated sports scheduling system
2. Player performance tracking and analysis
3. Training and match planning optimization
4. Data-driven decision support
5. Visual dashboards and reporting
6. Resource and time management optimization
7. Lightweight and scalable architecture.

Overall Architecture / Workflow

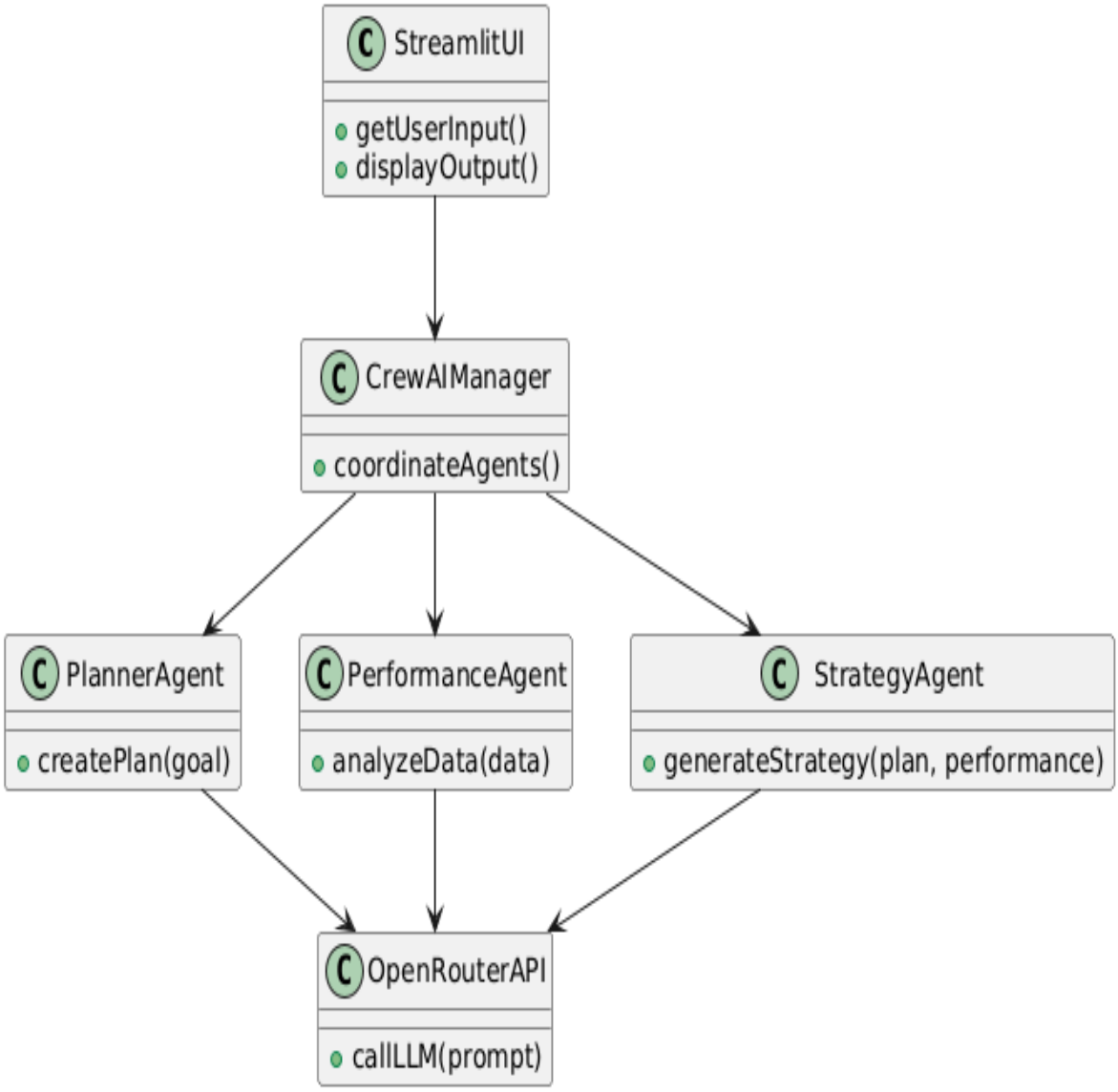
System Architecture Overview

System Workflow

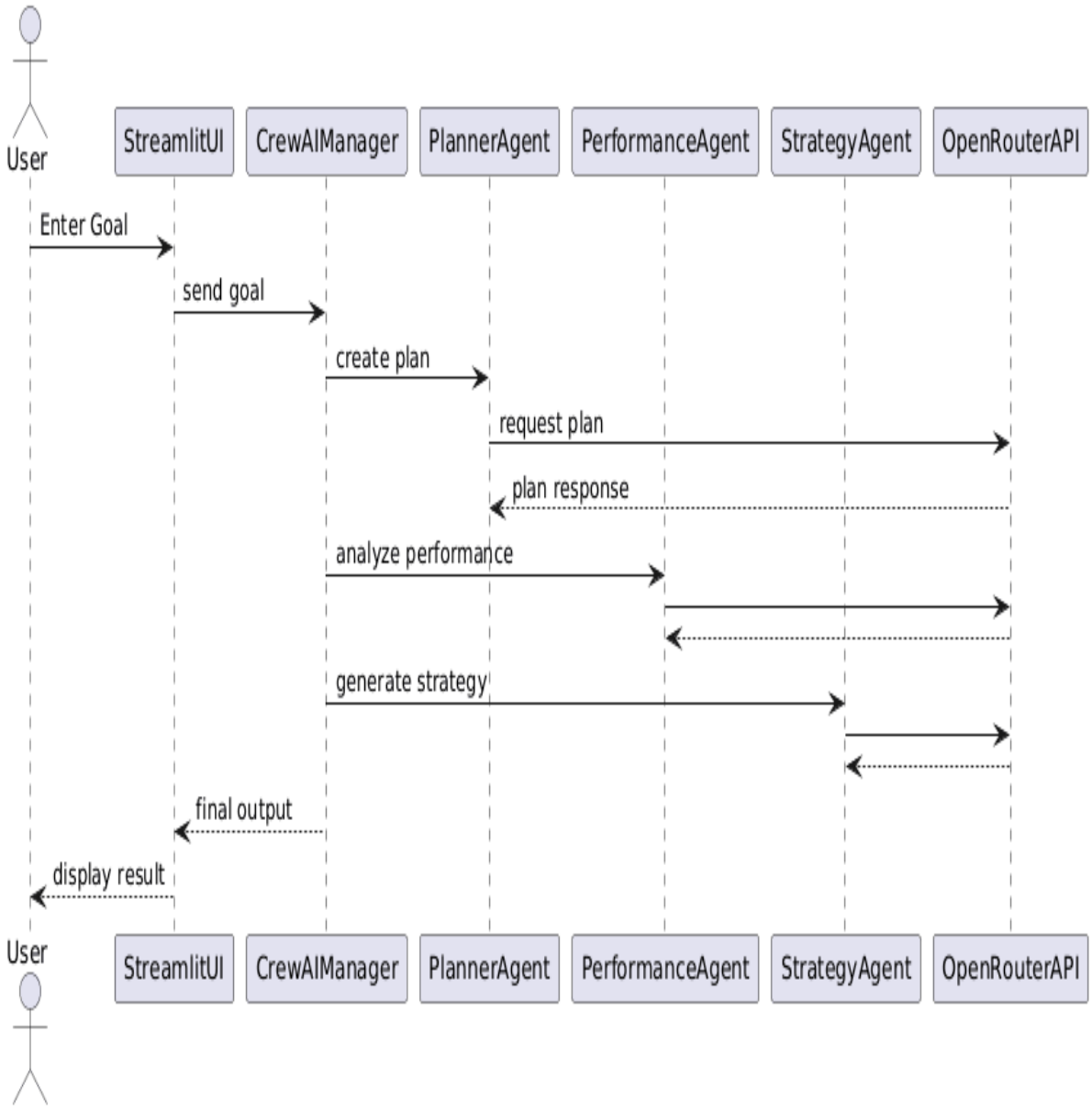
1. User opens the web application
2. User inputs player data and scheduling requirements
3. Backend validates the input
4. Data is processed and structured
5. Analytics module evaluates constraints and performance
6. Optimized schedules and recommendations are generated
7. Results are visualized in dashboards and reports
8. Output is displayed to the user

Diagrams

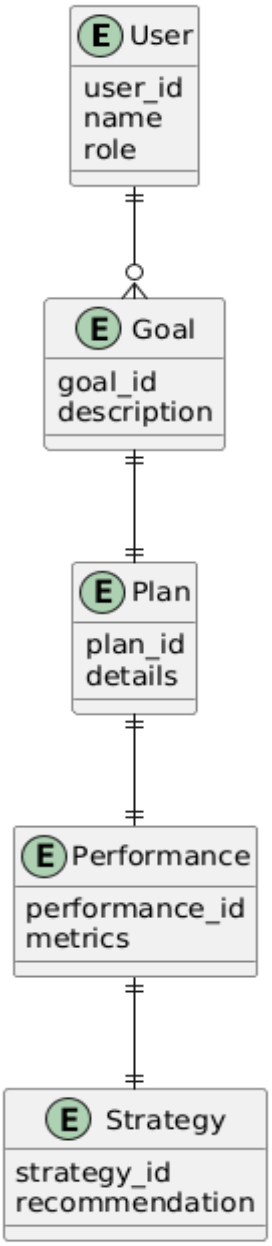
1. Class diagram



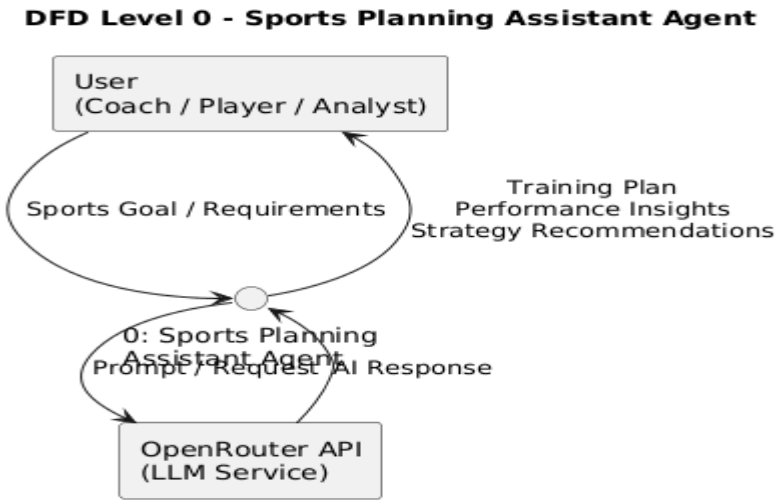
2.Sequence Diagram



3.ER Diagram


User
user_id
name
role
Goal
goal_id
description
Plan
plan_id
details
Performance
performance_id
metrics
Strategy
strategy_id
recommendation

4.Data Flow Diagram (Level 0)



System Design Details

Application Modules

a) User Interface Module

- Handles input forms and dashboards
- Displays schedules and analytics
- Maintains session state

b) Data Processing Module

- Cleans and structures input data
- Handles missing or inconsistent values

c) Planning & Optimization Module

- Generates efficient schedules
- Avoids conflicts and overlaps

d) Analytics Module

- Evaluates player performance
- Provides insights and trends

e) Reporting Module

- Generates visual reports
- Displays charts and summaries

Data Design

Since the system is AI-driven, heavy relational databases are not mandatory. However, lightweight storage is implemented for session and logging purposes.

- Logical Data Entities

  1. Player

     - player_id
     - name
     - age

- performance_stats

2. Schedule
   - schedule_id
   - date
   - activity_type
   - assigned_players
3. Performance Data
   - record_id
   - player_id
   - metrics
   - timestamp
4. System Logs
   - log_id
   - activity
   - status

Non-Functional Requirements Security Aspects

Security Aspects

- Secure data handling
- Input validation
- No sensitive data exposure

Performance Aspects

- Fast response time
- Efficient data processing
- Lightweight system design

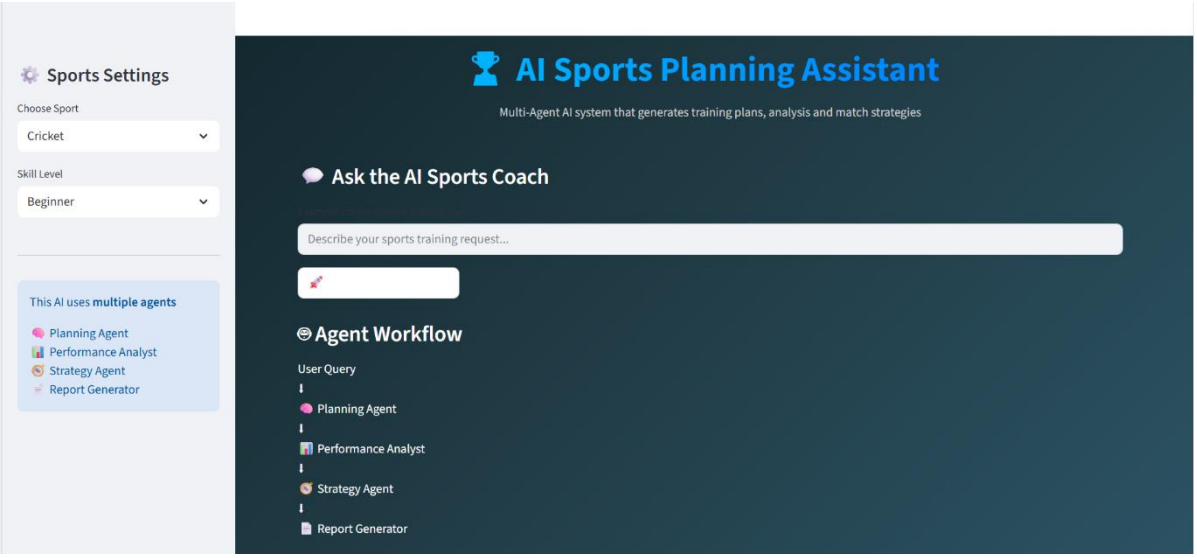
Tools & Technologies Used

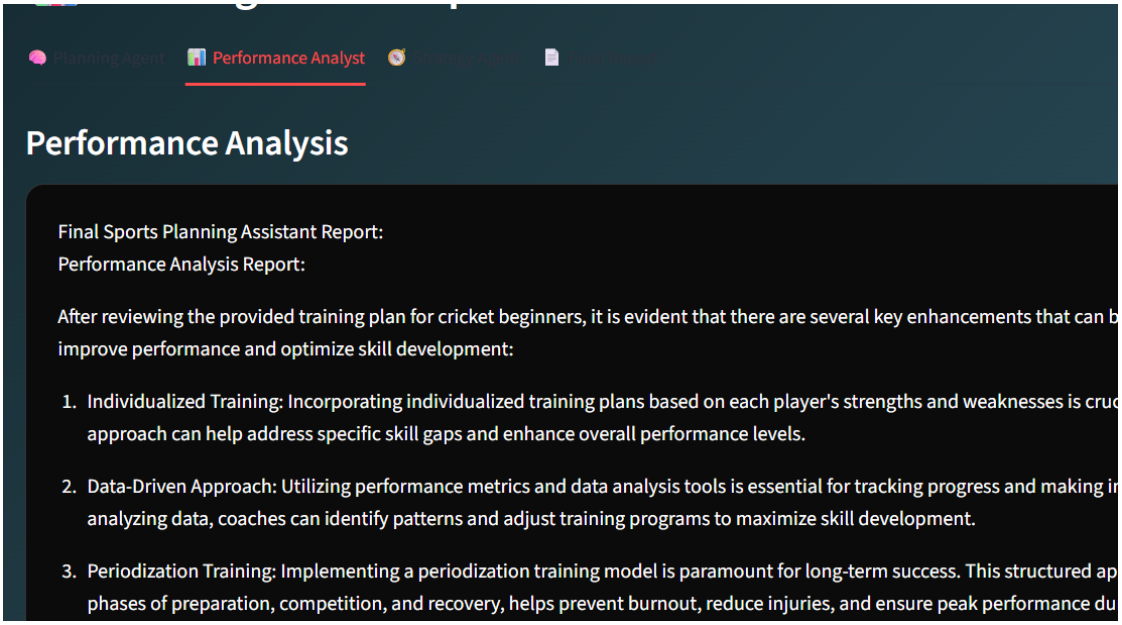
- Python (Core backend logic)
- Streamlit (Frontend UI)
- OpenRouter API (LLM Integration)
- JSON / SQLite (Optional storage)
- dotenv (Environment variable management)
- PlantUML (Diagram creation)
- Git (Version control)

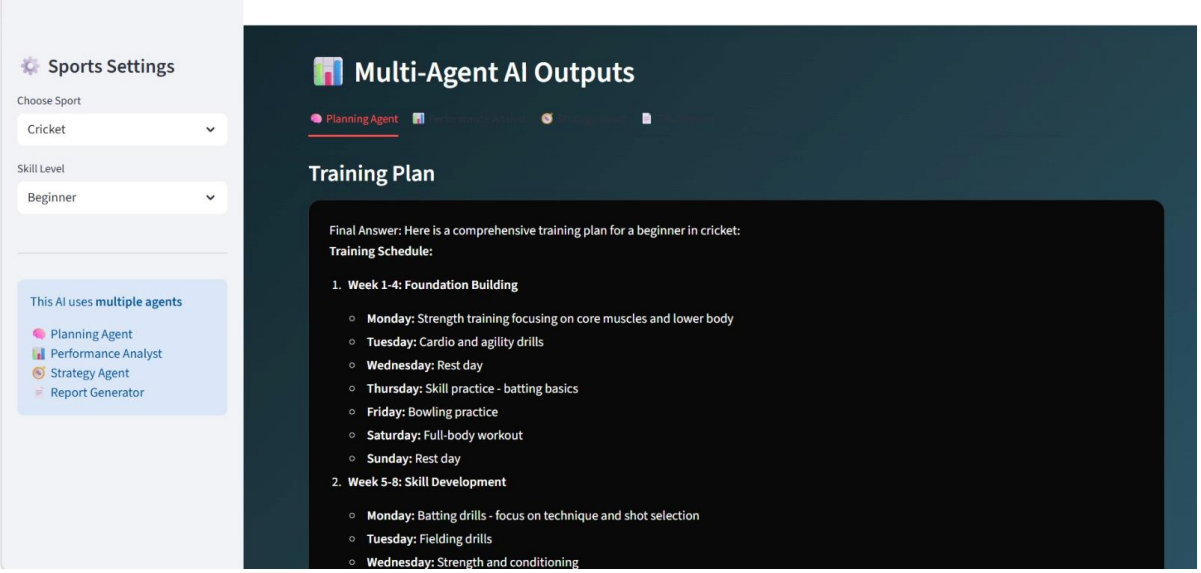
Results & Output

Screenshots / Outputs

- Sports AI Studio Dashboard




Performance Analyst
Performance Analysis
Final Sports Planning Assistant Report:
Performance Analysis Report:
After reviewing the provided training plan for cricket beginners, it is evident that there are several key enhancements that can b
improve performance and optimize skill development:
1. Individualized Training: Incorporating individualized training plans based on each player's strengths and weaknesses is cruc
approach can help address specific skill gaps and enhance overall performance levels.
2. Data-Driven Approach: Utilizing performance metrics and data analysis tools is essential for tracking progress and making i
analyzing data, coaches can identify patterns and adjust training programs to maximize skill development.
3. Periodization Training: Implementing a periodization training model is paramount for long-term success. This structured ap
phases of preparation, competition, and recovery, helps prevent burnout, reduce injuries, and ensure peak performance du


Sports Settings
Choose Sport
Cricket
Skill Level
Beginner
This AI uses multiple agents
Planning Agent
Performance Analyst
Strategy Agent
Report Generator
Multi-Agent AI Outputs
Planning Agent
Training Plan
Final Answer: Here is a comprehensive training plan for a beginner in cricket:
Training Schedule:
1. Week 1-4: Foundation Building
Monday: Strength training focusing on core muscles and lower body
Tuesday: Cardio and agility drills
Wednesday: Rest day
Thursday: Skill practice - batting basics
Friday: Bowling practice
Saturday: Full-body workout
Sunday: Rest day
2. Week 5-8: Skill Development
Monday: Batting drills - focus on technique and shot selection
Tuesday: Fielding drills
Wednesday: Strength and conditioning

Planning Agent | Performance Analyst | Strategy Agent | Final Report

Winning Strategy

Final Sports Planning Assistant Report:
Performance Analysis Report:

After reviewing the provided training plan for cricket beginners, it is evident that there are several key enhancements that can be made to further improve performance and optimize skill development:

1. Individualized Training: Incorporating individualized training plans based on each player's strengths and weaknesses is crucial. This tailored approach can help address specific skill gaps and enhance overall performance levels.
2. Data-Driven Approach: Utilizing performance metrics and data analysis tools is essential for tracking progress and making informed decisions. analyzing data, coaches can identify patterns and adjust training programs to maximize skill development.
3. Periodization Training: Implementing a periodization training model is paramount for long-term success. This structured approach, which incl phases of preparation, competition, and recovery, helps prevent burnout, reduce injuries, and ensure peak performance during important ma
4. Mental Skills Training: Integrating mental skills training into the program is essential for enhancing focus, concentration, and confidence.

Planning Agent | Performance Analyst | Strategy Agent | Final Report

Final AI Sports Report

Final Sports Planning Report

Cricket Training Plan for Beginners (4-Week Program)

Week 1: Fundamental Skills Development

Batting Technique (Days 1, 3, 5)

- **Day 1:** Focus on grip, stance, and basic shot execution.
- **Day 3:** Practice driving and defensive shots.
- **Day 5:** Work on playing lofted shots and footwork.

Bowling Accuracy (Days 2, 4, 6)

- **Day 2:** Concentrate on proper bowling action and release point.
- **Day 4:** Practice bowling line and length variations.

Key Outcomes

- Automated sports planning achieved
- Reduced manual scheduling effort significantly
- Improved decision-making using analytics
- Enhanced team coordination
- Built a scalable and modular system

Conclusion

The Sports Planning Assistant Agent demonstrates the effective use of data analytics in sports management. The system integrates user interface, backend processing, and analytics into a unified platform that improves efficiency and planning accuracy.

Through this project, valuable knowledge was gained in system design, data analysis, and real-world problem-solving. The project highlights how intelligent systems can transform traditional sports planning into a smart, data-driven process.

Future Scope & Enhancements

1. Integration with real-time sports data APIs
2. AI/ML-based predictive performance analysis
3. Mobile application development
4. Cloud deployment for scalability
5. Integration with wearable fitness devices
6. Advanced analytics dashboards
7. Multi-user and role-based access system